Rusi Wang

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RESEARCH INTERESTS

Transportation network modeling and optimization; multi-commodity flow; dynamic traffic assignment; multimodal transportation; travel demand management; congestion pricing.

EDUCATION

Tongji University Sep 2024–present
Ph.D. in Urban Mobility
Shanghai, China; College of Transportation; advisor: Prof. Chi Xie

Tongji University Sep 2021–May 2024
M.Sc.(Eng.) in Vehicle Operation Engineering
Shanghai, China; Institute of Rail Transit

University of Electronic Science and Technology of China Sep 2017–Jul 2021
B.Eng. in Internet of Things Engineering
Chengdu, China; School of Information and Communication Engineering

Institut Supérieur d'Électronique de Paris Sep 2019–Feb 2020
Visiting Undergraduate Student Awarded by China Scholarship Council (CSC)
Paris, France

National University of Kaohsiung Feb 2019–Jun 2019
Exchange Student
Kaohsiung, Taiwan, China; Department of Electrical Engineering

SELECTED PUBLICATIONS

Journal articles

1. Chi Xie, **Rusi Wang**, Dianlei Wang, Bo Zou, Xiaowen Fu, Xiqun Chen, and Qing-Chang Lu. Incorporating revenue loss and congestion cost into rail freight subsidy design: Lessons learned from the China-Europe freight transportation network. *Transport Policy*, 174:103819, 2025. [DOI]

Conference papers

1. **Rusi Wang** and Chi Xie. A convex programming model for dynamic ridesharing user equilibrium. Accepted for presentation at the *106th Annual Meeting of the Transportation Research Board*, Washington, D.C., 10–14 January 2027. Paper No. TRBAM-27-06071.
2. **Rusi Wang**, Chi Xie, Bo Zou, and Xiaowen Fu. Mitigating revenue loss and congestion surcharge by rail freight subsidy in a multimodal multicommodity freight transportation market. *105th Annual Meeting of the Transportation Research Board*, Washington, D.C., 11–15 January 2026. Paper No. TRBAM-26-04956. [Link]

Earlier publications are listed on my personal homepage.

WORKING PAPERS

- **Rusi Wang** and Chi Xie. A Nesterov-style convex programming model for dynamic user equilibrium with maximum entropy selection. *Working paper; planned journal submission*.
- **Rusi Wang** and Chi Xie. A convex programming model for dynamic user equilibrium based on Nesterov's network equilibrium model. *Working paper; planned conference submission*.

WORK IN PROGRESS

- **Rusi Wang** and Chi Xie. Dynamic equilibrium in multimodal networks with carpooling, autonomous taxis, and public transit. *Work in progress*.
- **Rusi Wang** and Chi Xie. Second-best congestion pricing in dynamic traffic networks: A bilevel optimization model with a convex dynamic user equilibrium formulation. *Work in progress*.

RESEARCH EXPERIENCE

Visiting undergraduate research, LISITE Lab, Institut Supérieur d'Électronique de Paris — Sep 2019–Feb 2020

- Studied melanoma classification with convolutional neural networks trained on dermoscopic images from the ISIC database, under Dr. Ercüment Yılmaz and Prof. Maria Trocan.

PROJECTS

Loopae, software for transportation network equilibrium analysis — 2026–present
Developed an interactive environment for computing, analyzing, and visualizing transportation network equilibria; continuing to refine and extend its capabilities.

PRESENTATION

A convex programming model for dynamic ridesharing user equilibrium. Upcoming: accepted for presentation at the 106th Annual Meeting of the Transportation Research Board, Washington, D.C., 10–14 January 2027.

SCHOLARSHIPS AND AWARDS

Chuangmai Scholarship, Tongji University — 2023
Excellent Student Scholarship, University of Electronic Science and Technology of China — 2019
Scholarship Under the State Scholarship Fund, China Scholarship Council — 2019
Honorable Mention, Interdisciplinary Contest in Modeling — 2019
First Prize, Sichuan Provincial Undergraduate Electronic Design Contest — 2018

ACADEMIC SERVICE

Journal reviewer, *Socio-Economic Planning Sciences* (2026).

Updated 16 October 2026.